Comprehensive Interrupt-Based Sense-Think-Act System

This project shows an Arduino Uno based Interrupt Based Smart Security System. Three digital sensors, two LEDs and a buzzer are used in the Sense Think Act model. It monitors security events and carries out recurring activities by combining Timer1 Interrupts with Pin Change Interrupts (PCI). The project's goal is to show real time system behavior and concurrent interrupt management.

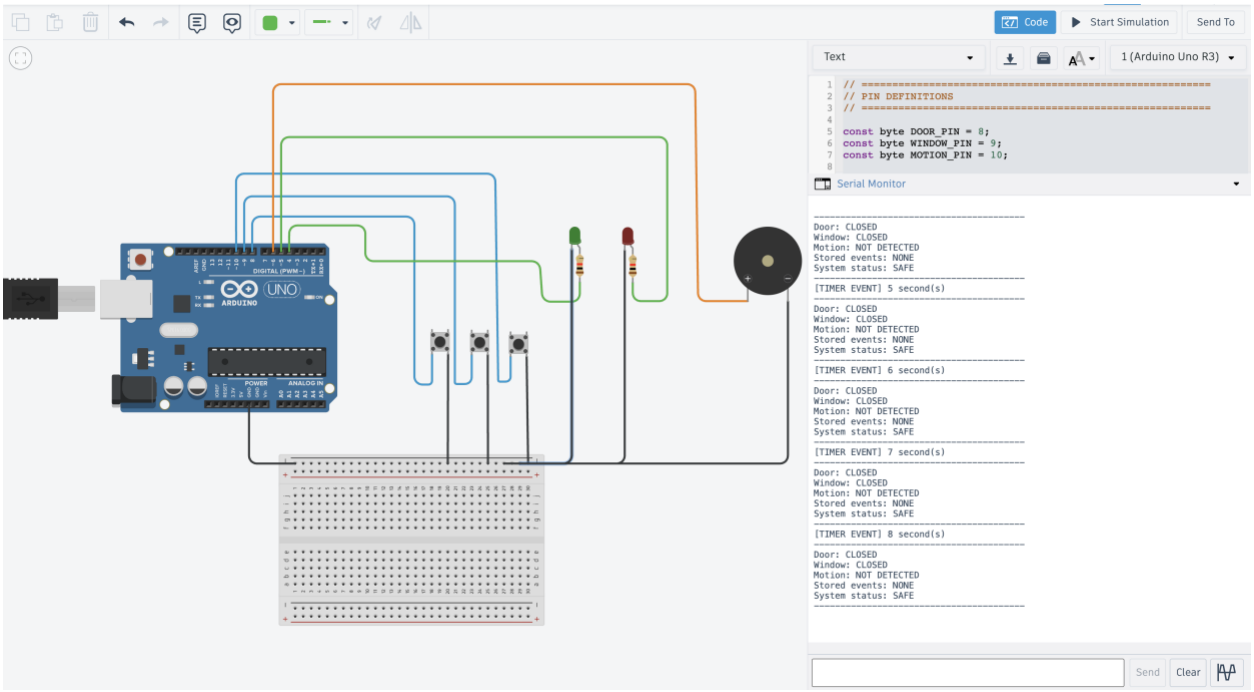
Three push buttons that stand in for door, window and motion sensors make up the system. An alarm situation is indicated by a red LED and buzzer but a green LED shows that the system is functioning normally. To configure hardware, read sensor inputs, execute system logic, update outputs and display data on the Serial Monitor, the software is separated into several functions. The Sense Think Act methodology is used throughout the programmed. The sensors first identify occurrences (Sense). The software then evaluates whether the device should stay in Safe mode, switch to Warning mode, or turn on Alarm mode (Think) based on the stored sensor states. Finally, the buzzer and LEDs react to the system status. The program is simpler to comprehend, test and maintain because to its modular design.

Digital pins D8, D9, and D10 are monitored via Pin Change Interrupts (PCI). An interrupt happens instantly and sets a flag for the main program to process whenever one of these sensor inputs changes. Timer1 is set up to produce an interrupt once every second. Each Interrupt Service Routine (ISR) simply sets flags and updates shared variables rather than carrying out lengthy tasks. The remaining processing is carried out by the main loop, which adheres to sound interrupt programming techniques. Periodic system updates and the blinking green status LED are handled by the timer interrupt, whilst real time sensor events are handled independently by the PCI.

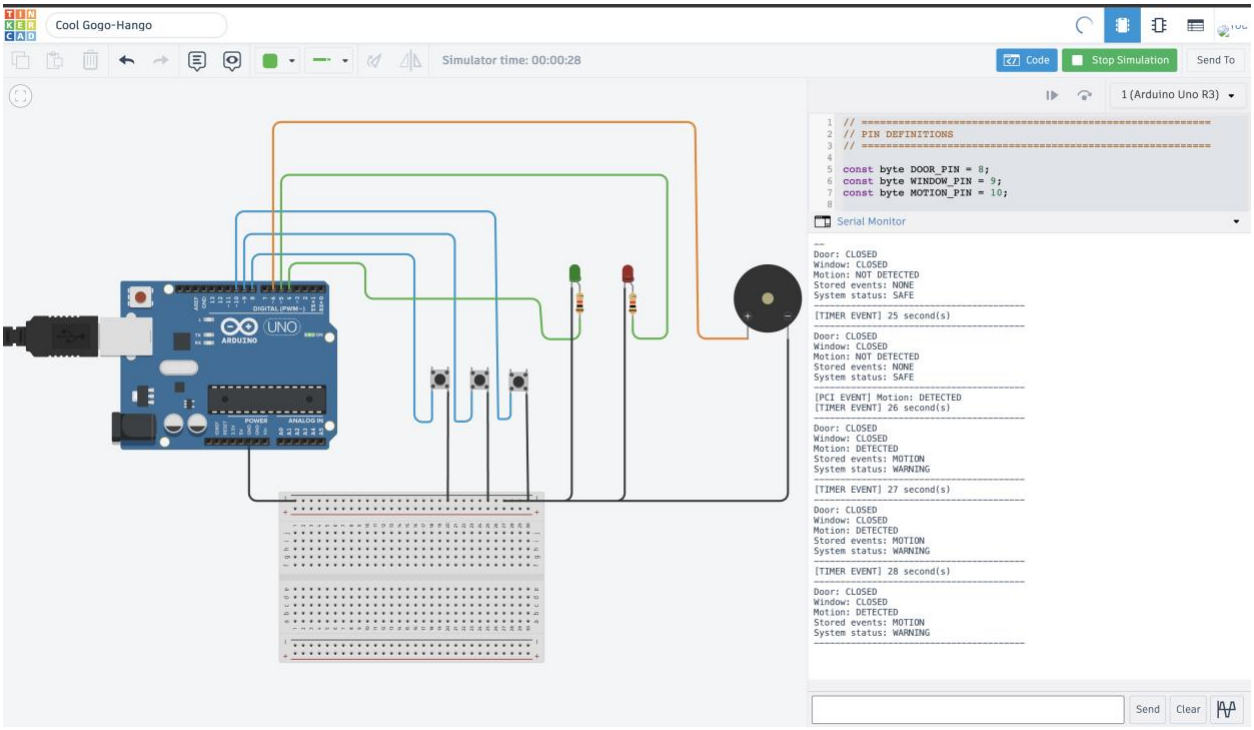
One of the challenges encountered during development was identifying two simultaneous button touches in Tinkercad. The original alarm logic did not always activate appropriately because it is challenging to press simulated push buttons and a mouse at the same time. The system was able to identify connected security incidents that took place within a three second span by temporarily storing sensor events. Button bounce, which resulted in frequent interrupt occurrences was another problem. By introducing a debounce delay prior to processing fresh sensor changes, this was reduced.

To sum up, the project effectively illustrates a real time interrupt driven embedded system. The system was able to react to security events effectively thanks to the usage of Pin Change Interrupts, Timer1 Interrupts, modular programming and the Sense Think Act architecture. By integrating periodic tasks with event driven processing in a dependable and well organized design, the final solution satisfies the Distinction requirements.

Screenshot 1



Screenshot 2



Screenshot 3

